Selling a Viral Product

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Abstract: Sellers often launch a new product with different price-quality packages (versions) when social learning is prevalent. Yet few papers explain the seller's versioning incentive from a social learning perspective. This paper explores when and how observational learning incentivizes a monopolist to sell different versions even in a common value setting. The findings highlight the pivotal role of consumers' private information quality in determining two distinct selling strategies. In markets with noisy private signals, the seller offers a single version; with precise signals, she adds a cheaper basic version.

Keywords: menu pricing, observational learning

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1 Introduction

Do firms profit from selling multiple versions when launching a new product? The economic literature has provided various explanations. However, most existing theories are static in nature, while in a digital era, new product markets often feature a dynamic observational learning process. When Apple launched the first iPhone in 2007, no one knew exactly how much value it would bring to everyday life. Over time, the public came to understand the value through the growing number of users. It is such public belief, rather than private tastes, that drives market demand in the long run. What is the optimal menu for the firm to launch such a novel product? When is it profitable to sell multiple versions, like iPhone and iPhone Pro, instead of a single version?

To address these questions, this paper proposes a dynamic model in which consumers share a common value but arrive sequentially with different information, and offers a new justification for the multi-version policy. In the presence of observational learning, a cheap basic version serves as a tool to guide social learning in the short run. It allows the first batch of consumers to try at a cheap price. A monopolist then gets to sell a much more expensive premium version, which would not have attracted any purchasers if it were the only option on the market. However, this role of versioning does not always work. When consumers have noisy private information, a single-version menu is optimal. Otherwise, it is profitable to add a basic version.

Reward-based crowdfunding exemplifies such new product markets. It is common practice for sellers to launch their crowdfunding campaigns with different price-quality packages[1] and consumers observe the number of purchases (backers) for each package. The multi-version policy is also widely adopted by firms selling online service products, such as Grammarly, ChatGPT, and D-ID Studio. Their motivation is precisely to let the cheap version go viral online. Once the market recognizes its core value, it will hopefully be ready for a premium version. With a simple and tractable model, this paper examines a monopoly pricing and versioning problem from the new Internet era, providing fresh insights into optimal selling strategies in various online markets.

A monopolist releases a new product for which consumers share an unknown binary common value. The seller aims to maximize expected long-run average profits. She can

[1] For instance, a comic book writer may offer both a digital version and a paperback version of their new book at different prices. Similarly, a game designer might sell a wide range of packages. Expensive ones often offer premium features to enhance the gaming experience.

offer a single-version or two-version menu with exogenous, observable vertical qualities and (fixed) prices[2]. An infinite number of consumers then arrive one at a time, each with a private signal about the value. They also observe their predecessors' decisions and obtain public information from there. At the end of each period, they either choose a version to buy or abstain. The learning part of the model is built on Smith and Sørensen (2000) and Herrera and Hörner (2012).

The article first examines the optimal price in a single-product benchmark model, focusing on the relationship between the optimal price, long-run demand elasticity and private signal informativeness. It then discusses the optimal learning patterns and when to offer two versions.

In the single-product benchmark, charging a higher price increases the margin (price effect) but reduces the probability of a buy cascade in the limit (quantity effect). The elasticity of demand depends on how informative consumers' private signals are. With a noisy signal, the quantity effect dominates and a lower price is optimal. To stay safe, the seller may even set a cheap price that triggers an immediate herd on the purchase decisions. A precise signal, however, weakens the quantity effect, incentivizing the seller to choose a high price. In doing so, she bets on the prior probability that the core product is good and reaps the fruit of learning as consumers become increasingly optimistic.

The two-version model introduces another layer of trade-off. Now the question boils down to whether to introduce a cheaper basic version. This is because, under the assumption of zero marginal production cost, the seller will always keep the premium product on the market.[3]

On the cost side, adding a cheaper basic version brings a conditional information rent. The seller must lower the premium version's price to prevent high-belief consumers from purchasing the basic version. Unlike in a standard screening model, this rent arises only along a good learning path.

The benefit of a basic version lies in facilitating the initial learning process. Suppose

[2]The fixed pricing assumption applies to many real-life situations. Many firms do not change prices frequently due to managerial inattention and brand image concerns (Arcidiacono et al. (2020), DellaVigna and Gentzkow (2019), Reimers and Waldfogel (2017), Phillips (2015)). Coca-Cola faced a huge backlash from consumers in 1999 when their then CEO Ivester considered introducing a new vending machine that changed price with the outside temperature. Most crowdfunding projects offer a static price for each package. Grammarly's pricing plan has remained unchanged for years.

[3]Imposing a higher production cost on the premium product only strengthens the seller's incentive to offer a basic version. The main results remain unchanged qualitatively.

consumers receive highly precise signals. As discussed in the benchmark case, the seller's top priority now is to charge a high price as long as consumers do not immediately run into a no-purchase cascade. A basic version helps here because it allows early consumers to try at low cost, generates information, and opens the door for later buyers to purchase the premium version at a price that would not have sold on its own.

A noisy signal, however, encourages the seller to stay safe by charging a cheap price on the premium version. Adding an even cheaper option only creates unnecessary information rents. As such, a single-version menu is optimal. Theorem 2 provides a threshold characterization of how signal informativeness determines the seller's decision to expand the menu.[4]

The theoretical framework provides a foundation for future empirical analysis of the interaction between observational learning and the optimal menu. It highlights a novel relationship between private information quality and selling strategy, which has been less studied in the literature. The next section provides a more detailed discussion.

2 Related literature

An important early paper that examines the monopoly (fixed) pricing problem with observational learning is Welch (1992). In contrast to my model, Welch assumes a uniformly distributed state and a finite number of agents. Moreover, his signal structure is relatively noisy. The posterior expected value updates so slowly that the issuer (seller) will underprice to completely avoid a rejection cascade. My paper adds to the literature by demonstrating that when the private signal becomes precise, it is optimal to charge a high price and risk a rejection cascade. In such cases, a multi-version menu yields even higher profits.

[4] As the main results highlight the role of private signal informativeness, it would be helpful to give some real-world examples where consumers may receive private information of varying precision. The consumer's private information is generally noisier if the core product is innovative. In the crowdfunding context, a consumer's private evaluation of the first book of a new series will not be as precise as their evaluation of the 10th book in the series. Another good example is the market for new treatments, as discussed in (Arieli, Koren, and Smorodinsky, 2022). A doctor gathers information by using limited free samples from pharmaceutical companies within the doctor's patient community. The realized success rate then gives each doctor a private signal about the value of the new treatment. The signal tends to be more precise if doctors receive more free samples, the samples are similar in effectiveness to the majority of products, or their patient communities are more diversified.

Bose, Orosel, Ottaviani, and Vesterlund (2006, 2008) investigate a dynamic pricing problem in the informational cascade setting with finite signals (binary in Bose et al. (2008)). Their setup bears similarities to my multi-version model. Because offering multiple versions with different prices at the outset seems to be a static alternative to dynamically adjusting the price of a single version. However, dynamic pricing offers the seller more flexibility to adjust their strategy as the learning process progresses. This paper offers new insights into the optimal menu design in this more constrained environment.

Several other studies have explored the optimal selling strategy in the presence of social experimentation. For instance, Bonatti (2011) investigates optimal dynamic menus for selling new experience goods to consumers with both a common value component and private taste. Laiho and Salmi (2021) considers a dynamic pricing problem when consumers can delay purchases. Bergemann and Välimäki (2002)[5] explores how dynamic competition among firms affects pricing and entry decisions. This literature assumes sales generate information via experimentation and simplifies the learning process to a publicly observable Brownian motion process. The assumption of public signal makes it difficult to discuss how private signal quality shapes the optimal selling strategy.

Another closely related field of study is menu pricing, which traces back to seminal works by Stigler (1963), Adams and Yellen (1976), Mussa and Rosen (1978) and Stokey (1979). Since then, numerous articles have explored conditions under which a monopolist prefers a multiple-item menu over a single-item menu (Salant, 1989; Anderson and Dana, 2009). A recent paper by Sandmann (2023) highlights the role of consumers' risk preferences. The evolving bundling literature (Haghpanah and Hartline, 2021; Ghili, 2023; Yang, 2024) establishes conditions for the optimality of pure and nested bundling. The conditions rely on relative values for and sold-alone quantities of different goods. This screening literature typically assumes consumers' willingness to pay is exogenous. In contrast, my paper features endogenously formed variations in consumers' preferences due to observational learning. It allows me to unravel a novel observation on versioning and private signal quality.

Many empirical papers have already explored observational learning in various contexts, such as kidney exchange markets (Zhang, 2010), microloan markets (Zhang and

[5]Other related papers include Bergemann and Välimäki (1997), Bergemann and Välimäki (2000), and Bergemann and Välimäki (2006)

Liu, 2012), music platforms (Newberry, 2016), and housing markets (Fan, Weng, Zhou, and Zhou, 2023). However, few have examined the effect of consumers' private information quality on the seller's pricing and versioning choices. According to Zhang and Liu (2012), investors in microloan markets behave differently during the learning process when they find their predecessor's private information is more precise. Based on this observation, the theoretical model developed here could guide further study of optimal selling strategies in such markets, especially when private information quality varies across different products.

3 Model

A monopolist seller plans to launch a new product. A countably infinite number of short-lived agents $t = 1, 2, \ldots$ with unit demand arrive one at a time.

State of the world. The product's core innovation is either a good match with the needs of the market, denoted by $\omega = 1$, or a poor match, denoted by $\omega = 0$. Neither the seller nor the agents observes the realized state. They share a common prior $\mu_1 := \Pr(\omega = 1) \in (0, 1)$.

Actions. The seller offers a static menu with up to two options: a premium version H with full features and a basic version with reduced features. I denote by $q_H \in (0, \infty)$ the premium quality, which can be interpreted as the amount of full features, and $q_L \in (0, q_H]$ the basic quality. These version qualities are publicly observable and exogenously given. The seller sets a pair of prices $(p_L, p_H) \in \mathbb{R}^2_+$ at the beginning of the game. In each period t, an agent buys a version, denoted by $a_t \in \{L, H\}$, or abstains, $a_t = r$.

Payoff. Agents receive a quasi-linear payoff from purchasing the new product: $u(q_a, \omega) - p_a$, for $a \in \{L, H\}$. The function $u : \mathbb{R} \times \{0, 1\} \to \mathbb{R}$ is twice continuously differentiable in q_a, strictly increasing, and strictly supermodular in (q_a, ω). The outside option ($a_t = r$) is normalized to give zero payoff.

The risk-neutral seller maximizes her long-run average profits

$$\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} [\mathbf{1}(a_t = L)p_L + \mathbf{1}(a_t = H)p_H].$$

Signal structure. Each agent, upon arrival, receives a private signal s_t about the state.

The signal is identically and independently distributed across t. Following Herrera and Hörner (2012), I assume without loss that the set of signal realizations, denoted by S, is either $[0,1]$, $\mathbb{R}_+$ or $\mathbb{R}$. Conditional on the state ω, the private signal follows a twice differentiable distribution $F(\cdot|\omega)$ which satisfies the strict Monotone Likelihood Ratio Property, that is, the signal likelihood ratio $\sigma(s) := \frac{f(s|1)}{f(s|0)}$ is strictly increasing on S. Let $\bar{\beta} := \sup_{s\in S}\{\sigma(s)\}$ and $\underline{\beta} := \inf_{s\in S}\{\sigma(s)\}$. The following analysis focuses on the case of bounded signals, i.e., $0 < \underline{\beta} < \bar{\beta} < \infty$. A private signal is *more informative* if $\underline{\beta}$ is lower or $\bar{\beta}$ is higher.

Timing. At the beginning of the game, Nature chooses the state ω. The seller announces the menu $\{(q_L, p_L), (q_H, p_H)\}$. In each period $t \in \{1, 2, ..., \infty\}$, an agent arrives. She observes the menu, decisions of previous agents, i.e., the public action history $h_t := (a_1, a_2, ..., a_{t-1})$, and a private signal s_t. She then chooses a product from the menu or walks away with nothing.

Equilibrium concept. The analysis focuses on pure-strategy perfect Bayesian equilibrium. In cases where an optimal menu does not exist[6], we will find a sequence of ε-optimal menus that converge to a well-defined limit menu. The comparative statics results are built on the associated limit profit maximization problem.

Definition 1 (ε-Optimal Menu)**.** *For any real non-negative number ε, an ε-optimal menu is one from which the seller cannot obtain more than ε in expected payoff by deviating.*

Discussion. The observable version qualities q_H and q_L differ from the unobserved match ω. Think of the iPhone and iPhone Pro: they share the core design features and operating system, but differ in battery life, storage, and screen size. If the seller can choose the basic quality, she will lower it as much as possible under mild conditions (e.g. multiplicative separable $u(\cdot)$). This is because adding features to the basic version only raises the information rent. So, q_L can be interpreted as an exogenous lower bound on basic quality.

The assumption of **supermodularity** means that the marginal value of a feature rises when the core innovation fits the market well. This assumption matches everyday experience. iPhone users care more about battery life when they like the core design. Gamers pay more attention to extra features when they enjoy the graphics and gameplay.

[6] Non-existence occurs when the seller prefers to offer two versions (Theorem 1) and a strict inequality constraint becomes binding.

The **static menu** assumption is motivated by the scale of online platforms. On Kickstarter or Steam, menus change rarely compared to the flow of new buyers. As discussed in the introduction, firms also have many other reasons to avoid frequent changes. This restriction on how often the firm can change the menu opens a venue for versioning to matter.

With **unbounded private signals**, versioning loses its role. Pricing and learning no longer interact, because long-run demand[7] equals the prior μ_1. The problem then reduces to a standard screening problem with a fraction μ_1 of high-value buyers and $1-\mu_1$ of low-value buyers. For simplicity, I assume **zero marginal production cost**. A constant marginal cost would not alter the qualitative results. Finally, the **tie-breaking rule** assumes buyers choose to purchase or pick the better version whenever indifferent. With an atomless belief distribution, this assumption is without loss.

4 Preliminary Analysis

To set up the seller's optimization problem, the first challenge lies in characterizing the long-run demand under different menus. This section shows how each menu induces a distinct learning process, through which public information accumulates over time. Section 4.2 then derives the long-run demand functions for different learning patterns. The proofs are collected in Appendix A.

4.1 Linking Prices to Learning Patterns

The learning process is captured by the evolution of the public likelihood ratio: $l_t := \frac{\Pr(\omega=1|h_t)}{\Pr(\omega=0|h_t)} \in [0, \infty]$ for any $t \in \mathbb{N}$ and public action history $h_t \in \{r, L, H\}^t$.[8] From Smith and Sørensen (2000), we can derive the transition probabilities and the continuation function which depends on the public likelihood ratio and the chosen action today. Denote the continuation function by $\varphi(a, l) := l\frac{\Pr(a|l,\omega=1)}{\Pr(a|l,\omega=0)}$.

Definition 2 (Public Likelihood Ratio Process)**.** *Assume $\omega = 0$. The public likelihood ratio process $\langle l_t \rangle_{t=1}^{\infty}$ is a Markov process with an initial state $l_1 = \frac{\mu_1}{1-\mu_1}$. The transition*

[7] The limit probability of a buy cascade.

[8] Under this definition, a higher public likelihood ratio indicates that the good state is more likely. Note that this differs from Smith and Sørensen (2000), which instead defines the ratio as the inverse, $\frac{\Pr(\omega=0|h_t)}{\Pr(\omega=1|h_t)}$.

probability function $\rho : [0,\infty] \times [0,\infty] \to [0,1]$ *satisfies, for any* $a \in \{r, L, H\}$ *with* $\Pr(a|l, \omega = 0) \neq 0$,

$$\rho(l, l') = \begin{cases} \Pr(a|l_t, \omega = 0), & \text{if } l' = \varphi(a, l) \\ 0, & o.w. \end{cases} \tag{1}$$

The state space of the public likelihood ratio can be divided into cascade sets and learning sets. A cascade occurs at a public likelihood ratio $l \in [0, \infty]$ if the agent chooses the same action regardless of her private signal. Otherwise, active learning happens. The rest of the section describes three learning patterns that can emerge under different menus, each featuring distinct learning dynamics.

To simplify the expressions, let

$$x_a := \frac{p_a - u(q_a, 0)}{u(q_a, 1) - p_a}, \forall a \in \{L, H\}, x_\Delta := \frac{p_H - p_L - (u(q_H, 0) - u(q_L, 0))}{(u(q_H, 1) - u(q_L, 1)) - (p_H - p_L)}.$$

These ratios increase in the prices and the price difference.

Definition 3 (Single-Premium Pattern)**.** *The public likelihood ratio process exhibits a single-premium pattern if* $\frac{p_H - u(q_H,0)}{u(q_H,1) - u(q_H,0)} \leq \frac{p_L - u(q_L,0)}{u(q_L,1) - u(q_L,0)}$.

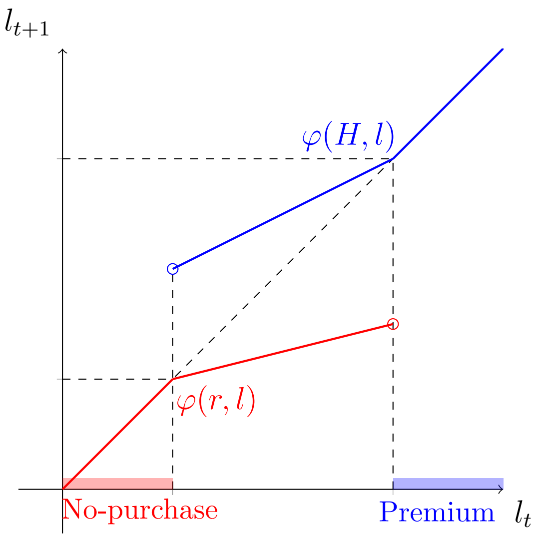
In the the single-premium pattern, the state space of the public likelihood ratio is partitioned into three sets: a no-purchase cascade set, $[0, \frac{x_H}{\bar{\beta}}]$, a learning set, $(\frac{x_H}{\bar{\beta}}, \frac{x_H}{\underline{\beta}})$, and a premium cascade set: $[\frac{x_H}{\underline{\beta}}, \infty]$.

Figure 1a shows the state space and the continuation function. Starting from the learning set, agents face two choices. Buying the premium version signals good news to future agents; not buying signals bad news. In the long run, the process converges to one of two outcomes: a purchase herd, in which the product goes viral, or a no-purchase herd, in which the launch fails.

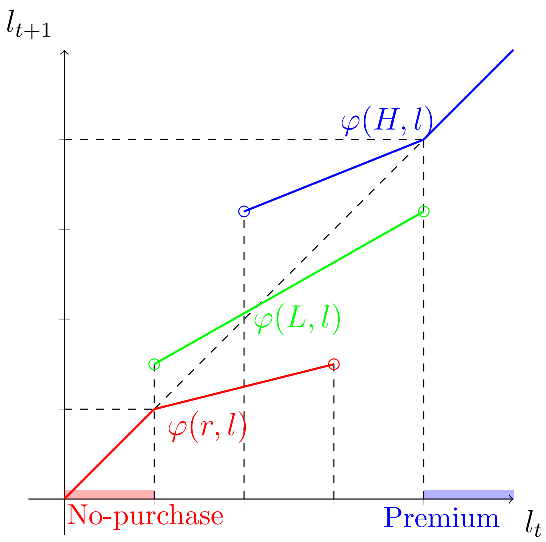
The next two patterns involve purchases of the basic version, which can play two distinct roles. In the first, the basic-short pattern, it supports social learning but only in the short run. In the second, the basic-long pattern, a buy cascade for the basic version can arise in the long run, guaranteeing the seller a base profit.

Definition 4 (Basic-Short Pattern)**.** *The public likelihood ratio process exhibits a basic-short pattern if* $\frac{p_L - u(q_L,0)}{u(q_L,1) - u(q_L,0)} < \frac{p_H - u(q_H,0)}{u(q_H,1) - u(q_H,0)} \leq 1$ *and* $\frac{x_\Delta}{\bar{\beta}} < \frac{x_L}{\underline{\beta}}$.

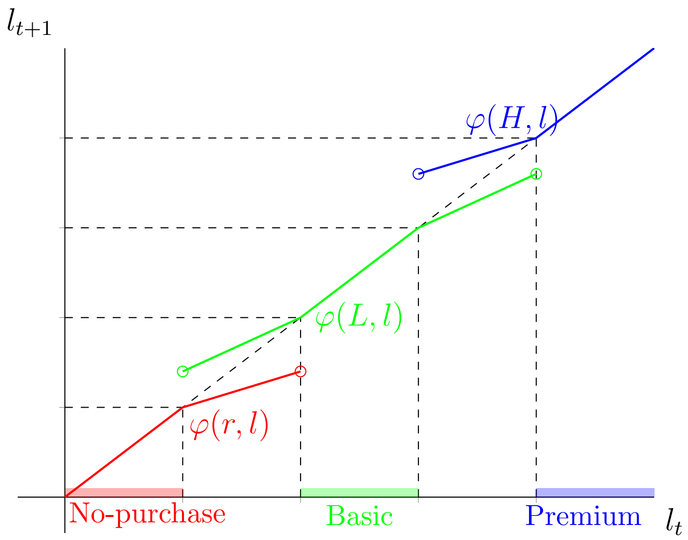
In the basic-short pattern, the state space is partitioned into five sets (see Figure 1b):



(a) Single-premium pattern



(b) Basic-short pattern



(c) Basic-long pattern

Figure 1: Three types of learning patterns

Note: the three figures plot continuation functions in different learning patterns. In the single-premium and basic-short patterns, we run into a premium cascade (blue shading area) when the public likelihood ratio is high; and a no-purchase cascade (red shading area) when the public likelihood ratio is low. If the two prices are far apart, a basic cascade set (green shading area) emerges in between as in the basic-long pattern.

1. No-purchase cascade set: $[0, \frac{x_L}{\bar{\beta}}]$
2. Slow learning set: $(\frac{x_L}{\bar{\beta}}, \frac{x_\Delta}{\bar{\beta}})$ in which agents choose between a basic version and no-purchase
3. Fast learning set: $(\frac{x_\Delta}{\bar{\beta}}, \frac{x_L}{\underline{\beta}})$ in which agents choose between a premium version, a basic version and no-purchase
4. Slow learning set: $(\frac{x_L}{\underline{\beta}}, \frac{x_\Delta}{\underline{\beta}})$ in which agents choose between a basic version and a premium version
5. Premium cascade set: $[\frac{x_\Delta}{\underline{\beta}}, \infty]$

In the fast learning set, agents choose among all three actions, which speeds up the learning process. Buying the basic version signals good news when public belief is low. As belief rises, the same action becomes weaker news and eventually bad news. As in the single-premium pattern, the long-run outcome is a "hit or miss": either a herd of premium purchases or a herd of no-purchase almost surely happens.

The pattern occurs under the strict inequality $x_\Delta \frac{1}{\bar{\beta}} < x_L \frac{1}{\underline{\beta}}$, which means the price gap is not too large. Otherwise, agents would herd on the basic version before trying out the premium version. As we will see later, the strict inequality creates a discontinuity in the long-run profit function, leading to an existence issue (Theorem 1).

Definition 5 (Basic-Long Pattern)**.** *The public likelihood ratio process exhibits a basic-long pattern if* $\frac{p_L - u(q_L,0)}{u(q_L,1)-u(q_L,0)} < \frac{p_H - u(q_H,0)}{u(q_H,1)-u(q_H,0)} \leq 1$ *and* $\frac{x_\Delta}{\bar{\beta}} \geq \frac{x_L}{\underline{\beta}}$*.*

The state space is partitioned into five sets (see Figure 1c):

1. No-purchase cascade set: $[0, \frac{x_L}{\bar{\beta}}]$
2. Basic learning set: $(\frac{x_L}{\bar{\beta}}, \frac{x_L}{\underline{\beta}})$ in which agents choose between a basic version and no-purchase
3. Basic cascade set: $[\frac{x_L}{\underline{\beta}}, \frac{x_\Delta}{\bar{\beta}}]$
4. Premium learning set: $(\frac{x_\Delta}{\bar{\beta}}, \frac{x_\Delta}{\underline{\beta}})$ in which agents choose between a basic version and a premium version
5. Premium cascade set: $[\frac{x_\Delta}{\underline{\beta}}, \infty]$

Unlike the basic-short pattern, the learning dynamics here involve only two actions. However, in the long run, a herd of basic purchases may arise. When the menu induces a learning process starting from the premium learning set, the basic version plays two roles: it signals bad news and it secures a long-run profit guarantee. In this way, the seller obtains some downside protection, since even along a bad learning path all buyers converge to purchasing the basic version.

4.2 Demand: Long-Run Cascade Probability

When the seller maximizes the long-run average profit, what happens in finite time does not matter. The average number of purchases for each version will converge to the probability of a purchase cascade for that version in the limit. Lemma 1 and its corollary provide a closed-form formula for the long-run demand in each learning pattern. To derive it, I impose the following assumption on the signal distributions.

Assumption 1. *For each $\omega \in \{0, 1\}$, $F(\cdot|\omega)$ satisfies*

1. *(Weak) Increasing Hazard Ratio Property: the hazard ratio $\frac{1-F(s|0)}{1-F(s|1)}\frac{f(s|1)}{f(s|0)}$ is weakly increasing on S.*

2. *(Weak) Increasing Failure Ratio Property: the failure ratio $\frac{F(s|0)}{F(s|1)}\frac{f(s|1)}{f(s|0)}$ is weakly increasing on S.*

When agents will possibly take only two actions, the two properties together ensure that if an agent begins more optimistic, she stays more optimistic after seeing what others have bought. Formally, the continuation function $\varphi(a, l)$ is *weakly* increasing in l, the public likelihood ratio today (Proposition 1 and 2 Herrera and Hörner (2012)). The assumption is not a knife-edge case. A broad family of signal distributions satisfies both properties. Examples include the linear density functions studied in Smith and Sørensen (2000) (Section 3.B) and Herrera and Hörner (2012) (Section 3.2.1).[9] Smith et al. (2021) offers an alternative characterization: (weak) monotonicity holds if the density function of $\ln(\frac{s}{1-s})$ is weakly log-concave.

This monotone updating has a sharp implication: the public likelihood ratio will converge to a boundary of the learning set rather than jumping into the interior of the

[9]The exponential and power distributions also qualify, but are not bounded. Most signal distributions used in economics satisfy the Increasing Hazard Ratio Property, including the exponential, normal, power and Gamma distributions; see Sections 4.7 of Herrera and Hörner (2012).

cascade sets. This is also true in the case where agents will possibly take three actions. Because the updated public belief after a middle action, i.e. purchasing the basic version L, is always in between those after more extreme actions, i.e. no purchase r or purchasing the premium version H. The following lemma states the result formally.

Lemma 1. *Assume Assumption 1 holds. Let Y be a learning set. $\lim_{t\to\infty} l_t \in \partial Y$.*

Because the public likelihood ratio process is a conditional martingale, the expectation of its limit equals its starting point. Together with Lemma 1, we can derive a neat expression for the probability of cascades and, in turn, for long-run demand. For any $a \in \{L, H\}$, let $\lambda(a) := \lim_{t\to\infty} \frac{1}{t}\sum_{s=1}^{t} \mathbf{1}(a_s = a)$.

Corollary 1. *Assume Assumption 1 holds.*

1. *In the single-premium pattern, $\lambda(H) = \frac{1}{1+l_1}(\underline{\beta} + x_H)\frac{\bar{\beta}l_1 - x_H}{\bar{\beta}x_H - \underline{\beta}x_H}$ and $\lambda(L) = 0$;*
2. *In the basic-short pattern, $\lambda(H) = \frac{1}{1+l_1}(\underline{\beta} + x_\Delta)\frac{\bar{\beta}l_1 - x_L}{\bar{\beta}x_\Delta - \underline{\beta}x_L}$ and $\lambda(L) = 0$;*
3. *In the basic-long pattern, $\lambda(H) = \frac{1}{1+l_1}(\underline{\beta} + x_\Delta)\frac{\bar{\beta}l_1 - x_\Delta}{\bar{\beta}x_\Delta - \underline{\beta}x_\Delta}$ and $\lambda(L) = 1 - \lambda(H)$ if the process starts from the premium learning set; $\lambda(H) = 0$ and $\lambda(L) = \frac{1}{1+l_1}(\underline{\beta} + x_L)\frac{\bar{\beta}l_1 - x_L}{\bar{\beta}x_L - \underline{\beta}x_L}$ if the process starts from the basic learning set.*

5 The Optimal Selling Strategy

What distinguishes this paper from earlier work on menu pricing is that the menu shapes demand by changing the underlying learning structure. This section begins with a single-product benchmark. The benchmark clarifies how signal informativeness links learning to long-run demand elasticity, laying the foundation for understanding when and why observational learning motivates the seller to offer two versions. We then turn to the optimal menu and identify new economic forces in the seller's trade-off between a single- and two-version menu. The proofs are collected in Appendix B.

5.1 A Single Product Benchmark

Suppose the seller offers only the full version (p_H, q_H). Similar to standard monopoly pricing problems, the seller's expected long-run profit equals the product of the ex-ante purchase cascade probability $\lambda_H(x_H)$ and the price $p_H(x_H)$. Raising the price shifts the learning boundaries downward. Hence, the learning process starts closer to a no-purchase cascade, lowering long-run demand.

Demand becomes less responsive to price when signals are more informative, that is, when the best and worst signal realization become more convincing. Intuitively, highly informed agents are harder to sway: a worst-signal agent is unlikely to buy, and a best-signal agent is unlikely to abstain. Because manipulating the learning outcome is so costly, the seller gives up securing a high purchase cascade probability. She instead favors a risky, high price and bet on a good learning path to yield large expected profits. The following proposition states the result formally.

Proposition 1 (Demand Elasticity)**.** *The long-run demand elasticity* $|\frac{d\ln\lambda_H}{d\ln p_H}|$ *monotonically increases in* $\bar{\beta}$ *and decreases in* $\underline{\beta}$*.*

5.2 Optimal Menu and Learning Structure

Section 4.1 illustrates that from the perspective of observational learning, adding a cheap basic version can serve two purposes: (a) guiding learning in the short run, or (b) guaranteeing a base profit.

The following discussion explains why the profit-guarantee role is often suboptimal.

Assumption 2. *The utility function* $u(q_L, \omega)$ *satisfies*

$$\frac{\partial}{\partial q_L} \ln \frac{u(q_L, 0)}{u(q_L, 1) - u(q_L, 0)} \geq 0 \tag{2}$$

or

$$u(q_L, 0) = 0 \tag{3}$$

In words, condition (2) says that a one-percent increase in basic quality raises utility from the basic version in the bad state by more (in percentage terms) than it raises the basic version's utility gap across states. A large class of utility functions satisfy this assumption, including any multiplicatively separable $u(q, \omega)$.

Lemma 2. *Assume Assumption 1 and 2 holds. The basic-long pattern is suboptimal.*

First, a market that sells the basic version only is never optimal. By selling a premium version instead, the seller faces the same demand curve but earns a higher profit margin from each unit because the premium quality is higher $q_H > q_L$. In the basic-long pattern, starting below the premium learning set ends with only the basic version sold, and therefore cannot be optimal.

What is the trade-off in switching from a single-premium menu to a two-version menu with a basic-long pattern that begins from the premium learning set? The two-version menu secures a base profit from the basic cascade. Yet if public belief rises, the seller must pay information rents to high-belief premium buyers to discourage them away from the basic version.

Under Assumption 2, such a switch is never optimal for any $q_L < q_H$. Both the base profit and the expected information rent increase in the basic quality. The assumption makes sure that the gain from a profit guarantee rises faster than the rent. As a result, the profit gap between the optimal two-version menu and the single-premium menu can cross zero at most once and only from below. However, as q_L approaches q_H, the information rent becomes so large that most profits come from the basic version buyers. The seller is then effectively launching straight into a basic cascade. A single-premium menu that sparks an immediate premium cascade yields strictly higher profit. Hence, the switch is never profitable.

The rest of the section discusses the existence issue and characterizes the optimal learning structure.

Theorem 1. *Assume Assumption 1 and 2 holds. An ε-optimal menu exists. It induces a learning process that feature binary actions. That is, as $\varepsilon \to 0$, the set of public beliefs at which agents will choose among three actions shrinks to empty.*

Both results in Theorem 1 relate to the fundamental reason why a two-version menu with a basic-short learning pattern can outperform a single-premium menu.

On the benefit side, a cheap basic version "postpones" an immediate no-purchase cascade. This allows the seller to charge a high premium price that would have caused an immediate no-purchase cascade if the premium version were the only product in the market. A two-version, basic-short menu effectively stretches the learning set towards lower beliefs, so the first several low-belief agents can try the new product at a low cost. The seller then extracts more surplus from high-belief buyers by lifting the premium price without advancing the onset of a no-purchase cascade. As illustrated by Figure 2, this learning benefit grows as the seller targets higher prices (lower cascade probabilities).

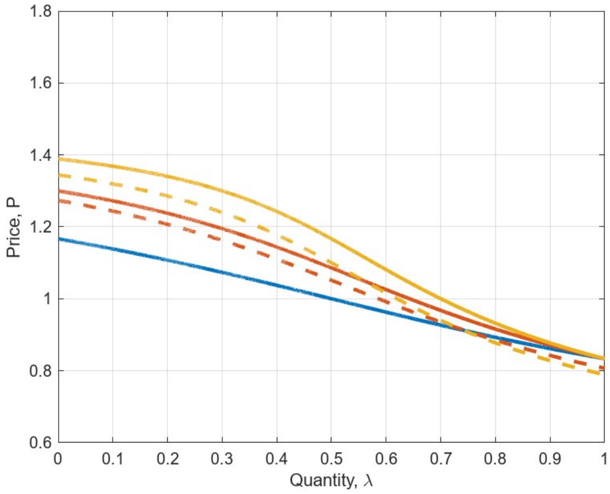


Figure 2: The long-run demand curves with and without information rent

Note: the x-axis is the long-run demand, i.e., the limit cascade probability of the premium purchase. The y-axis refers to the premium version's price. The utility function is $u(q_a, \omega) = q_a \cdot V_\omega, \forall a \in \{L, H\}$*, and the parameters are:* $q_H = 0.5, V_0 = 1, V_1 = 3, l_1 = 1, \bar{\beta} = 2, \underline{\beta} = 0.5$*. The blue curve plots the long-run demand curve with a single-premium menu* ($\frac{x_\Delta}{x_L} = 1$)*. For* $q_L = 0$*, the red solid curve plots the long-run demand curve with a two-version, basic-short menu and a smaller price difference between two versions,* $\frac{x_\Delta}{x_L} = 2$*. The yellow solid curve plots the long-run demand curve with a two-version, basic-short menu and a larger price difference between two versions,* $\frac{x_\Delta}{x_L} = 4$*. The dashed curves plots the long-run demand curves for* $q_L = 0.1$*. I used MATLAB to plot the figure.*

On the cost side, the size of the information rent varies endogenously and it affects long-run profit only along a good learning path. If the seller targets a small long-run demand, the probability of such a path is low, so the cost is minor. If she targets a large long-run demand, the rent outweighs the benefit. For instance, in the extreme, when the target demand is one (an immediate premium cascade), there is no learning and a cheap basic version only creates a sizable information rent.

As such, the long-run demand curve rotates clockwise as the seller widens the price gap (Figure 2). To capture the full learning benefit, the seller pushes the gap as wide as possible, which stretches the learning set. As the price gap approaches the upper bound, the fast learning set, where agents choose among all three actions, nearly disappears. Once the price gap *reaches* the upper bound, the learning process switches immediately to the basic-long pattern. Because of this pattern switch, the profit jumps discontinuous and an optimal two-version, basic-short menu does not exist. However, we can always find an ε-optimal menu.

The next result compares the premium cascade thresholds under the optimal two-version, basic short menu and the optimal single-premium menu. A key implication is that successful products tend to have higher expected quality compared to the single-product benchmark.

Lemma 3. *Assume Assumption 1 and 2 holds. The ε-optimal two-version menu induces a higher premium cascade threshold than that in the single-product benchmark.*

5.3 When to Offer Two Versions

Theorem 2 presents a threshold property of how signal informativeness affects the optimal number of menu options. The seller introduces a cheaper basic version if and only if private signals are sufficiently informative, in particular, when the most convincing bad news is strong enough.

Theorem 2. *Assume Assumption 1 and 2 holds. For any $\bar{\beta} > 0$, there exists a threshold $\underline{\beta}^* \in (0, \bar{\beta})$ such that a two-version menu is optimal if and only if $\underline{\beta} \leq \underline{\beta}^*$.*

From the single-product benchmark, we know that more informative signals make demand less responsive. Hence, the seller wants to raise the premium price rather than cut it to insure against a no-purchase cascade. This strong incentive to target a high price, and hence low demand, makes it profitable to use a basic version to guide social learning. With noisier signals, by contrast, the seller prefers a safer option. If a low price is the target in the first place, adding a basic version yields little benefit. Intuitively, there is no point in offering an even cheaper option if the seller already sells an affordable premium product.

6 Concluding Remarks

This paper studies when a two-version menu is optimal in new product markets with observational learning. In this setting, a cheaper basic version can serve two roles. First, it may secure a base profit if a bad learning path triggers a basic-purchase cascade. Second, it may facilitate learning at the outset while disappearing in the long run: by postponing a no-purchase cascade, the basic version lets the seller charge a higher premium price and capture more surplus once the market recognizes the product's value. Lemma 2 shows that the profit-guarantee role never works when the seller maximizes long-run profit.

Another unique insight of the paper is that the informativeness of private signals shapes the elasticity of long-run demand, and hence the seller's pricing strategy. With noisy signals, the seller launches the new product with a single premium version. With precise signals, the learning role of the basic version becomes effective, making a two-version menu optimal.

From an applied perspective, the conditions identified in Lemma 2 and Theorem 2 yield sharp predictions about when we will see more versioning of the kind that aims to facilitate observational learning, beyond its more familiar role of screening heterogeneous consumers.

Appendices

Appendix A Preliminary Analysis

A.1 Agent's Problem and Cascade Sets

The expected payoff for agent t, given prices (p_L, p_H), a public action history h_t and a private signal s_t is

$$\begin{cases} (1 - \Pr(\omega = 1|h_t, s_t))u(q_L, 0) + \Pr(\omega = 1|h_t, s_t)u(q_L, 1) - p_L, & a_t = L \\ (1 - \Pr(\omega = 1|h_t, s_t))u(q_H, 0) + \Pr(\omega = 1|h_t, s_t)u(q_H, 1) - p_H, & a_t = H \\ 0, & a_t = r \end{cases}$$

where $\Pr(\omega = 1|h_t, s_t)$ denotes the posterior belief of agent t after observing previous agents' actions and the private signal.

The seller effectively excludes a version from the market if the version's price is unreasonably high. For instance, when $\frac{p_L - u(q_L,0)}{u(q_L,1)-u(q_L,0)} \leq 1 < \frac{p_H - u(q_H,0)}{u(q_H,1)-u(q_H,0)}$, agents only consider the basic version. Since the marginal production cost is zero, the total surplus from selling the premium version is greater than that from the basic version at any posterior belief. This strategy is obviously sub-optimal for the seller. As another example, when $\frac{p_H - u(q_H,0)}{u(q_H,1)-u(q_H,0)} \leq \frac{p_L - u(q_L,0)}{u(q_L,1)-u(q_L,0)}$, the basic version becomes so expensive that no one would ever consider it. This results in agents behaving as they would in a single-product world.

A two-version market occurs under a menu with $\frac{p_H - u(q_H,0)}{u(q_H,1)-u(q_H,0)} < \frac{p_L - u(q_L,0)}{u(q_L,1)-u(q_L,0)} \leq 1$. Agents may choose whether to buy the basic version at first and then consider the premium version as the posterior belief increases.

To derive the cascade sets, we can write the posterior belief as

$$\begin{aligned} \Pr(\omega = 1|h_t, s_t) &= \frac{\Pr(s_t|h_t, \omega = 1)\Pr(\omega = 1, h_t)}{\sum_{\omega \in \{0,1\}} \Pr(s_t|h_t, \omega)\Pr(\omega, h_t)} = \frac{\Pr(s_t|\omega = 1)\Pr(\omega = 1, h_t)}{\sum_{\omega \in \{0,1\}} \Pr(s_t|\omega)\Pr(\omega, h_t)} \\ &= \frac{\sigma(s_t)\frac{\Pr(\omega=1,h_t)}{\Pr(\omega=0,h_t)}}{\sigma(s_t)\frac{\Pr(\omega=1,h_t)}{\Pr(\omega=0,h_t)} + 1} = \frac{\sigma(s_t)l_t}{\sigma(s_t)l_t + 1}. \end{aligned}$$

Following Smith and Sørensen (2000), it is easy to derive the cascade thresholds in Section 4. Take the premium cascade threshold in the single-premium pattern for example. An agent will purchase the premium version if and only if $\Pr(\omega = 1|h_t, s_t) = \frac{\sigma(s_t)l_t}{\sigma(s_t)l_t+1} \geq \frac{p_H-u(q_H,0)}{u(q_H,1)-u(q_H,0)}$. Rearrange the inequality and we have $\sigma(s_t) \geq \frac{x_H}{l_t}$. Then the agent will purchase the premium version regardless of her private signal if $\frac{x_H}{l_t} = \underline{\beta}$. Hence, the premium cascade threshold is $l_t = \frac{x_H}{\underline{\beta}}$.

A.2 Proof of Lemma 1 and Corollary 1

The proof relies heavily on the results from Smith and Sørensen (2000) and Herrera and Hörner (2012). To start with, let me discuss belief convergence and action convergence. The next result follows immediately from Lemma 3 and Theorem 1 in Smith and Sørensen (2000).

Lemma 4. *Assume $\omega = 0$. The public likelihood process $\langle l_t \rangle_{t=1}^{\infty}$ is a Markov process. The public likelihood ratio converges almost surely to a random variable $l_\infty = \lim_{t\to\infty} l_t \in [0, \infty)$. A cascade will happen in the limit.*

Assume $\omega = 1$. The inverse public likelihood ratio process $\langle \frac{1}{l_t} \rangle_{t=1}^{\infty}$ is a Markov process. The inverse public likelihood ratio converges almost surely to $\frac{1}{l_\infty} \in [0, \infty)$. A cascade will happen in the limit.

The corollary of Lemma 3 in Smith and Sørensen (2000) shows action convergence almost surely obtains if the private signal distribution is atomless. Combined with the claim above, it is easy to see that $\frac{1}{t}\sum_{s=1}^{t} \mathbf{1}(a_s = a)$ converges almost surely to $\Pr(l_\infty \in J_a)$ where J_a denotes the cascade sets for action a.

Next, we prove that the public likelihood ratio never jumps outside the learning set from which the process starts. A similar argument works for the inverse public likelihood ratio. For the case of two actions, Proposition 2 in Herrera and Hörner (2012) implies that the continuation functions $\varphi(a, l)$ is strictly increasing under the two properties in Assumption 1. Now we prove this is also true when the agents can take three actions.

Suppose that the process starts from a learning set (l_{min}, l_{max}) in which agents will possibly choose all three actions. Then, $l_{min} = \frac{x_L}{\beta}$ and $l_{max=} = \frac{x_\Delta}{\underline{\beta}}$. The continuation functions are

$$\varphi(H,l) = l\frac{1-F(\bar{s}(l)|1)}{1-F(\bar{s}(l)|0)}$$
$$\varphi(L,l) = l\frac{F(\bar{s}(l)|1)-F(\underline{s}(l)|1)}{F(\bar{s}(l)|0)-F(\underline{s}(l)|0)}$$
$$\varphi(r,l) = l\frac{F(\underline{s}(l)|1)}{F(\underline{s}(l)|0)}$$

where $\bar{s}(l) = \sigma^{-1}(x_\Delta/l)$ and $\underline{s}(l) = \sigma^{-1}(x_L/l)$. Assumption 1 implies that, for all $l \in [l_{min}, l_{max}]$, (1) $\varphi(r,l) \in [l_{min}, l_{max}]$ and (2) $\varphi(H,l) \in [l_{min}, l_{max}]$.

Intuitively, a belief update following the middle action should be smaller than that from extreme actions. Indeed, the following statement is true:

$$\frac{F(s''|1)-F(s'|1)}{F(s''|0)-F(s'|0)} > \frac{F(s'|1)-F(s|1)}{F(s'|0)-F(s|0)}, \forall s < s' < s''. \tag{4}$$

To see why, note that $F(s''|0)-F(s'|0) = \int_{s'}^{s''} \frac{f(s|1)}{\sigma(s)} dF(\cdot|1) < \frac{1}{\sigma(s')}(F(s''|1)-F(s'|1))$ under the Monotone Likelihood Ratio Property. Similarly, $F(s'|0) - F(s|0) > \frac{1}{\sigma(s')}(F(s'|1) - F(s|1))$. It follows that $\frac{F(s''|1)-F(s'|1)}{F(s''|0)-F(s'|0)} > \frac{1}{\sigma(s')} > \frac{F(s'|1)-F(s|1)}{F(s'|0)-F(s|0)}$. Therefore, $l_{min} \leq \varphi(r,l) \leq \varphi(L,l) \leq \varphi(H,l) \leq l_{max}$.

Since the public likelihood ratio never jumps outside the learning set, the limit likelihood ratio l_∞ must be one of the two boundaries of the adjacent cascade sets. By the Dominated Convergence Theorem, $E(l_\infty|\omega) = l_0$ for each $\omega \in \{0,1\}$. The cascade probabilities $\Pr(l_\infty \in J_a)$, and thus the long-run demand, can be easily derived from there.

Appendix B The Optimal Selling Strategy

B.1 Proof of Proposition 1

For any $x_H > 0$, $\ln \lambda_H(x_H) = -\ln(1+l_1) + \ln(\underline{\beta} + x_H) + \ln(\bar{\beta} l_1 - x_H) - \ln(\bar{\beta} - \underline{\beta}) - \ln x_H$. $\ln p_H(x_H) = \ln(u_{H0} + \frac{\bar{u}_H x_H}{1+x_H})$ which strictly increases in x_H. Let $(\ln p_H)^{-1}(\ln p)$ denote the inverse function of $\ln p_H(x_H)$. Then $|\frac{d \ln \lambda_H}{d \ln p}| = |\frac{\partial \ln \lambda_H}{\partial x_H}((\ln p_H)^{-1}(\ln p))||\frac{\partial (\ln p_H)^{-1}}{\partial \ln p}(\ln p)|$. Since $\ln p_H(x_H)$ is independent of $\bar{\beta}$ and $\underline{\beta}$, the bounds of the signal likelihood ratio, and strictly increasing in x_H, the demand elasticity decreases in signal informativeness if and only if $|\frac{\partial \ln \lambda_H}{\partial x_H}|$ decreases in $\bar{\beta}$ and increases in $\underline{\beta}$ for any $x_H > 0$. We can easily verify that by checking the partial derivative.

B.2 The Optimal Menu

Designing the menu under observational learning amounts to two choices: (a) which learning pattern to induce and (b) where to start the process.

Some pairs of learning patterns and initial positions can be easily ruled out. For instance, starting below the basic cascade set in the basic-long pattern is strictly dominated by a single-premium pattern. The former ends in either a no-purchase or basic cascade, while the latter results in either a no-purchase or premium cascade. But a premium cascade is more profitable than a basic cascade as selling a premium version with a higher version quality generates a larger total surplus.

For the remaining cases, the seller's problem can be written as follows. To further simplify notation, let $\mathbf{x}$ denote (x_L, x_Δ), and $\forall a \in \{L, H\}, u_{a\omega} := u(q_a, \omega), \bar{u}_a = u_{a1} - u_{a0}$. For each pattern, I also add two extra constraints because it is weakly suboptimal for the seller to increase the price if the process already starts with an immediate "bad" cascade (no-purchase or basic), or decrease the price if it starts with an immediate premium cascade.

1. Single-premium pattern:[10]

$$\max_{x_\Delta \geq 0} \quad \frac{1}{1+l_1}(\underline{\beta} + x_\Delta)\frac{\bar{\beta}l_1 - x_\Delta}{\bar{\beta}x_\Delta - \underline{\beta}x_\Delta}(u_{H0} + \bar{u}_H \frac{x_\Delta}{1+x_\Delta}) \tag{5}$$
$$\text{s.t. } x_\Delta = x_L, l_1\underline{\beta} \leq x_\Delta \leq l_1\bar{\beta}$$

2. Basic-short pattern:

$$\max_{x_L, x_\Delta \geq 0} \quad \frac{1}{1+l_1}(\underline{\beta} + x_\Delta)\frac{\bar{\beta}l_1 - x_L}{\bar{\beta}x_\Delta - \underline{\beta}x_L}(u_{H0} + \frac{\bar{u}_L x_L}{1+x_L} + \frac{(\bar{u}_H - \bar{u}_L)x_\Delta}{1+x_\Delta}) \tag{6}$$
$$\text{s.t. } x_\Delta > x_L, \bar{\beta}x_L > \underline{\beta}x_\Delta, x_\Delta \geq l_1\underline{\beta} \text{ and } l_1\bar{\beta} \geq x_L$$

[10]Note that $x_H = x_\Delta = x_L$ if $x_\Delta = x_L$. Alternatively, we can change the choice variable to x_H or x_L.

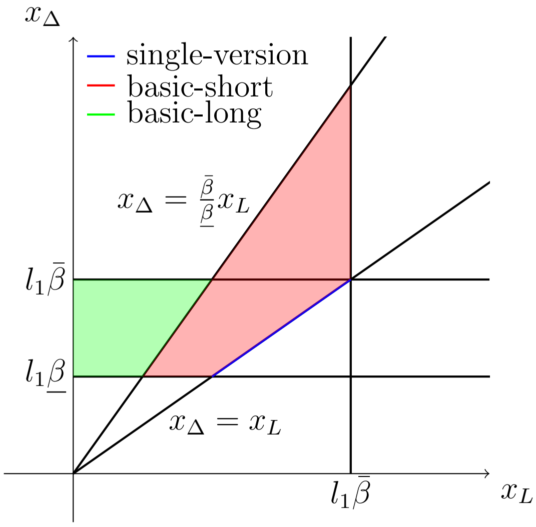


Figure 3: Three learning patterns in (x_L, x_Δ) space.

3. Basic-long pattern (starting from the premium learning set):

$$\begin{aligned}\max_{x_L, x_\Delta \geq 0} \quad & u_{L0} + \frac{\bar u_L x_L}{1+x_L} + \\ & \frac{1}{1+l_1}(\underline\beta + x_\Delta)\frac{\bar\beta l_1 - x_\Delta}{\bar\beta x_\Delta - \underline\beta x_\Delta}(u_{H0} - u_{L0} + (\bar u_H - \bar u_L)\frac{x_\Delta}{1+x_\Delta}) \qquad (7) \\ \text{s.t. } & x_\Delta > x_L, \bar\beta x_L \leq \underline\beta x_\Delta, l_1\underline\beta \leq x_\Delta \leq l_1\bar\beta\end{aligned}$$

Figure 3 plots the constraint sets of the three learning patterns in the space of (x_L, x_Δ). $x_\Delta = x_L$ represents the single-premium menu. $x_\Delta = \frac{\bar\beta}{\underline\beta} x_L$ separates the basic-short and basic-long pattern.

B.2.1 Proof of Lemma 2

As mentioned before, it is never optimal to start from below the premium learning set in the basic-long pattern. We now show that starting from the premium learning set is not optimal either.

Notice first that the constraint $(\bar\beta x_L \leq \underline\beta x_\Delta)$ must bind in the basic-long problem (7). Otherwise, the seller can always increase her payoff by increasing x_L without violating any other constraints. Substituting the binding constraint into the objective yields the following optimization problem:

$$\begin{aligned}\max_{x_\Delta} v_{2L}(x_\Delta; q_L) :=& u_{L0} + \frac{\bar{u}_L \frac{\beta}{\bar{\beta}} x_\Delta}{1 + \frac{\beta}{\bar{\beta}} x_\Delta} + \\ & \frac{1}{1+l_1}(\underline{\beta} + x_\Delta)\frac{\bar{\beta} l_1 - x_\Delta}{\bar{\beta} x_\Delta - \underline{\beta} x_\Delta}(u_{H0} - u_{L0} + (\bar{u}_H - \bar{u}_L)\frac{x_\Delta}{1+x_\Delta}) \\ & \text{subject to } l_1 \underline{\beta} \leq x_\Delta \leq l_1 \bar{\beta}\end{aligned} \tag{8}$$

"2L" in the subscript stands for the basic-long pattern. Let $x^*_{2L}(q_L)$ denote the solution to this program and $v^*_{2L}(q_L)$ its value function. Similarly, let $v_1(x_\Delta)$ denote the objective function in the single-premium problem (5), with an optimizer x^*_1 and value function v^*_1. Denote the difference between the two profit functions by

$$D(x_\Delta; q_L) := v_{2L}(x_\Delta; q_L) - v_1(x_\Delta) = u_{L0} A(x_\Delta) + \bar{u}_L B(x_\Delta) \tag{9}$$

where $A(x_\Delta) := 1 - \frac{1}{1+l_1}(\underline{\beta} + x_\Delta)\frac{\bar{\beta} l_1 - x_\Delta}{\bar{\beta} x_\Delta - \underline{\beta} x_\Delta}$ and $B(x_\Delta) := \frac{\frac{\beta}{\bar{\beta}} x_\Delta}{1 + \frac{\beta}{\bar{\beta}} x_\Delta} - \frac{1}{1+l_1}(\underline{\beta} + x_\Delta)\frac{\bar{\beta} l_1 - x_\Delta}{\bar{\beta} x_\Delta - \underline{\beta} x_\Delta}\frac{x_\Delta}{1+x_\Delta}$.

Since the basic quality does not affect the single-premium profit function, the difference between the *optimal* profits is $D^*(q_L) := D(x^*_{2L}(q_L); q_L)$, which is twice continuously differentiable on $q_L \in [0, q_H]$. We aim to show that $D(x^*_{2L}(q_L); q_L) < 0$ for all $q_L \in (0, q_H]$ under condition (2) in Assumption 2, i.e., $u(q_L, 0) > 0$ and $\frac{\partial}{\partial q_L} \ln \frac{u(q_L, 0)}{u(q_L, 1) - u(q_L, 0)} \geq 0$ for all $q_L \in (0, q_H]$. A similar argument works for the second condition (3), $u(q_L, 0) = 0$.

Step 1: we will show that $D(x_\Delta, q_L) = 0 \Rightarrow \frac{\partial D}{\partial q_L}(x_\Delta, q_L) > 0$ for all $(x_\Delta, q_L) \in \mathbb{R}^2_{++}$.

Assume $D(x_\Delta, q_L) = 0$. Then, $u_{L0} A(x_\Delta) + \bar{u}_L B(x_\Delta) = 0$ and we have

$$\frac{\partial D}{\partial q_L}(x_\Delta, q_L) = \frac{\partial u_{L0}}{\partial q_L} A - \frac{\partial \bar{u}_L}{\partial q_L}\frac{u_{L0}}{\bar{u}_L} A = A u_{L0}[\frac{\frac{\partial u_{L0}}{\partial q_L}}{u_{L0}} - \frac{\frac{\partial \bar{u}_L}{\partial q_L}}{\bar{u}_L}] = A u_{L0} \frac{\partial}{\partial q_L} \ln \frac{u_{L0}}{\bar{u}_L} \geq 0$$

The last inequality follows from our condition (2). Thus, the difference function $D^*(q_L) = D(x^*_{2L}(q_L); q_L)$ crosses zero at most once and only from below

Step 2: we will show that as q_L approaches the upper bound q_H, the optimal profit from the basic-long pattern is strictly lower than that from the single-premium pattern.

For any $x_\Delta \leq l_1 \bar{\beta}$, $\lim_{q_L \to q_H} v_{2L}(x_\Delta, q_L) = u_{H0} + \bar{u}_H \frac{\frac{\beta}{\bar{\beta}} x_\Delta}{1 + \frac{\beta}{\bar{\beta}} x_\Delta}$. So $\lim_{q_L \to q_H - 0} x^*_{2L}(q_L) = l_1 \bar{\beta}$. It follows that $\lim_{q_L \to q_H - 0} \frac{\partial D}{\partial q_L}(x^*_{2L}(q_L), q_L) = \frac{\partial u_{L0}}{\partial q_L} A(x^*_{2L}(q_L)) + \frac{\partial \bar{u}_L}{\partial q_L} B(x^*_{2L}(q_L)) > 0$

because the long-run demand (for the premium version) in the basic-long pattern, i.e., $\frac{1}{1+l_1}(\underline{\beta}+x_{2L}^*)\frac{\bar{\beta}l_1-x_{2L}^*}{\bar{\beta}x_{2L}^*-\underline{\beta}x_{2L}^*}$, approaches zero. Moreover, $\lim_{q_L\to q_H-0} v_{2L}^*(q_L)=u_{H0}+\bar{u}_H\frac{l_1\beta}{1+l_1\underline{\beta}}=v_1(l_1\underline{\beta})\leq v_1^*$. As a result, there exists a $\delta>0$ such that $D(x_{2L}^*(q_L);q_L)<0$ for all $q_L\in(q_H-\delta,q_H)$.

Since the function approaches q_H from below, Step 1 implies that $D(x_{2L}^*(q_L);q_L)<0$ for any $q_L\in[0,q_H]$.

B.2.2 Proof of Theorem 1

Now that we rule out the basic-long pattern, we can merge the single-premium and the basic-short problem into a general problem. Note also that choosing $x_L=l_1\bar{\beta}$ leads to zero demand which is never optimal. It is thus without loss to maximize the logarithm of the profit function:

$$\max_{x_L,x_\Delta\geq 0}\quad \ln v(x_L,x_\Delta):=\ln[\frac{1}{1+l_1}(\underline{\beta}+x_\Delta)\frac{\bar{\beta}l_1-x_L}{\bar{\beta}x_\Delta-\underline{\beta}x_L}(u_{H0}+\frac{\bar{u}_Lx_L}{1+x_L}+\frac{(\bar{u}_H-\bar{u}_L)x_\Delta}{1+x_\Delta})]$$
$$\text{s.t. } x_\Delta\geq x_L, \bar{\beta}x_L>\underline{\beta}x_\Delta, x_\Delta\geq l_1\underline{\beta} \text{ and } l_1\bar{\beta}>x_L \tag{10}$$

The optimal solution has a "bang-bang" structure in the dimension of x_Δ. To see why, the first-order partial derivative of $\ln v$ with respect to x_Δ is

$$\begin{aligned}
&\frac{1}{\underline{\beta}+x_\Delta}-\frac{\bar{\beta}}{\bar{\beta}x_\Delta-\underline{\beta}x_L}+\frac{(\bar{u}_H-\bar{u}_L)/(1+x_\Delta)^2}{u_{H0}+\frac{\bar{u}_Lx_L}{1+x_L}+\frac{(\bar{u}_H-\bar{u}_L)x_\Delta}{1+x_\Delta}}\\
&=\frac{-\underline{\beta}(x_L+\bar{\beta})}{(\underline{\beta}+x_\Delta)(\bar{\beta}x_\Delta-\underline{\beta}x_L)}+\frac{(\bar{u}_H-\bar{u}_L)/x_\Delta(1+x_\Delta))}{u_{H0}(\frac{1}{x_\Delta}+1)+\frac{\bar{u}_Lx_L}{1+x_L}(\frac{1}{x_\Delta}+1)+(\bar{u}_H-\bar{u}_L)}\\
&=\frac{1}{x_\Delta(1+x_\Delta)}[\frac{-\underline{\beta}(x_L+\bar{\beta})x_\Delta(1+x_\Delta)}{(\underline{\beta}+x_\Delta)(\bar{\beta}x_\Delta-\underline{\beta}x_L)}+\frac{\bar{u}_H-\bar{u}_L}{u_{H0}(\frac{1}{x_\Delta}+1)+\frac{\bar{u}_Lx_L}{1+x_L}(\frac{1}{x_\Delta}+1)+(\bar{u}_H-\bar{u}_L)}]
\end{aligned}\tag{11}$$

It is easy to verify that the term in square brackets in (11) strictly increases in x_Δ within the constraint set. Fixing any $x_L>0$, $\frac{\partial\ln v}{\partial x_\Delta}(x_L,x_\Delta)\geq 0$ thus implies $\frac{\partial\ln v}{\partial x_\Delta}(x_L,x_\Delta)$ increases in x_Δ. So $\ln v$ either increases in x_Δ or decreases first and then increases in x_Δ within the constraint set. The optimal x_Δ must either hits the constraint $x_\Delta\geq x_L$, which means that a single-premium menu is optimal, or stays as close as possible to $\bar{\beta}x_L>\underline{\beta}x_\Delta$, which means a two-version, basic-short menu is optimal.

When a single-premium menu is optimal, a unique optimal menu exists. When a two-version, basic-short menu is optimal, we can find an ε-optimal menu. Let $(x^*_{2S}, \frac{\underline{\beta}}{\bar{\beta}}x^*_{2S})$ denote the maximizer to the basic-short problem (6) on a relaxed constraint set where the constraint $\bar{\beta}x_L > \underline{\beta}x_\Delta$ is allowed to bind[11]. By the definition of the first-order partial derivative, for any $\varepsilon > 0$, there exists $\delta(\varepsilon) > 0$ such that

$$|\frac{v(x^*_{2S}, x_\Delta) - v(x^*_{2S}, \frac{\underline{\beta}}{\bar{\beta}}x^*_{2S})}{x_\Delta - \frac{\underline{\beta}}{\bar{\beta}}x^*_{2S}} - \frac{\partial v}{\partial x_\Delta}(x^*_{2S}, \frac{\underline{\beta}}{\bar{\beta}}x^*_{2S})| < \varepsilon, \forall x_\Delta \in (\frac{\underline{\beta}}{\bar{\beta}}x^*_{2S}, \frac{\underline{\beta}}{\bar{\beta}}x^*_{2S} + \delta(\varepsilon)) \tag{12}$$

which implies

$$|v(x^*_{2S}, x_\Delta) - v(x^*_{2S}, \frac{\underline{\beta}}{\bar{\beta}}x^*_{2S})| < (\varepsilon + |\frac{\partial v}{\partial x_\Delta}(x^*_{2S}, \frac{\underline{\beta}}{\bar{\beta}}x^*_{2S})|)|x_\Delta - \frac{\underline{\beta}}{\bar{\beta}}x^*_{2S}|, \forall x_\Delta \in (\frac{\underline{\beta}}{\bar{\beta}}x^*_{2S}, \frac{\underline{\beta}}{\bar{\beta}}x^*_{2S} + \delta(\varepsilon)).$$

Hence, $(x^*_{2S}, \frac{\underline{\beta}}{\bar{\beta}}x^*_{2S} + d)$ where $d = \min\{\delta(\varepsilon), \frac{1}{2}, \frac{\varepsilon}{2|\frac{\partial v}{\partial x_\Delta}(x^*_{2S}, \frac{\underline{\beta}}{\bar{\beta}}x^*_{2S})|}\}$ is an ε-optimal menu[12]. As $\varepsilon \to 0$, the fast learning set in which agents will possibly choose all three actions shrinks to empty because $\bar{\beta}x_L$ gets as close as possible to $\underline{\beta}x_\Delta$.

B.2.3 Proof of Lemma 3

First, we prove that $\frac{\partial \ln v}{\partial x_L}$ decreases in x_L and x_Δ within the constraint set in problem (10).

$$\begin{aligned}\frac{\partial \ln v}{\partial x_L} =& \frac{-1}{\bar{\beta}l_1 - x_L} - \frac{-\underline{\beta}}{\bar{\beta}x_\Delta - \underline{\beta}x_L} + \frac{\bar{u}_L}{(1+x_L)^2}\frac{1}{u_{H0} + \frac{\bar{u}_L x_L}{1+x_L} + \frac{(\bar{u}_H - \bar{u}_L)x_\Delta}{1+x_\Delta}} \\ =& \frac{-\bar{\beta}(x_\Delta - l_1\underline{\beta})}{(\bar{\beta}l_1 - x_L)(\bar{\beta}x_\Delta - \underline{\beta}x_L)} + \frac{\bar{u}_L}{(1+x_L)^2}\frac{1}{u_{H0} + \frac{\bar{u}_L x_L}{1+x_L} + \frac{(\bar{u}_H - \bar{u}_L)x_\Delta}{1+x_\Delta}}\end{aligned}$$

which is obviously decreasing in x_L on the constraint set. Moreover, $\frac{\partial^2 \ln v}{\partial x_L \partial x_\Delta} = \frac{-\bar{\beta}\underline{\beta}}{\left(\bar{\beta}x_L - \underline{\beta}x_0\right)^2} - \frac{\bar{u}_L}{(1+x_L)^2}\frac{(\bar{u}_H - \bar{u}_L)}{(1+x_\Delta)^2}\frac{1}{\left(u_{H0} + \frac{\bar{u}_L x_L}{1+x_L} + \frac{(\bar{u}_H - \bar{u}_L)x_\Delta}{1+x_\Delta}\right)^2} < 0$.

Let $(x^{2S}_L, x^{2S}_\Delta)$ be an optimal two-version menu that solves the basic-short problem (6) on the relaxed constraint set and (x^1_L, x^1_Δ) be an optimal single-version menu that solves the single-premium problem (5). Suppose for contradiction that $x^{2S}_\Delta \leq x^1_\Delta$. Then $x^1_L = x^1_\Delta \geq x^2_\Delta = \frac{\bar{\beta}}{\underline{\beta}}x^2_L > x^2_L$. From Figure 3, we know that the optimality of the two-version, basic-short menu implies $\frac{\partial \ln v}{\partial x_L}(x^{2S}_L, x^{2S}_\Delta) \leq 0$. Since $\frac{\partial^2 \ln v}{\partial x_L \partial x_\Delta} < 0$, $\frac{\partial \ln v}{\partial x_L}(x^{2S}_L, x^1_\Delta) <$

[11] The solution to this relaxed problem exists because the problem has a continuous objective function and a bounded, convex constraint set.

[12] d is well-defined since $\frac{\partial v}{\partial x_\Delta}(x_L, x_\Delta)$ never approaches infinity on the relaxed constraint set of (x_L, x_Δ).

$\frac{\partial \ln v}{\partial x_L}(x_L^{2S}, x_\Delta^{2S}) \leq 0$. Then, $\frac{\partial \ln v}{\partial x_L}(x_L^1, x_\Delta^1) \leq \frac{\partial \ln v}{\partial x_L}(x_L^{2S}, x_\Delta^1) < 0$ because $\frac{\partial \ln v}{\partial x_L}$ decreases in x_L. Together with the shape of the constraint set in Figure 3, $\frac{\partial \ln v}{\partial x_L}(x_L^1, x_\Delta^1) < 0$ suggests that (x_L^1, x_Δ^1) cannot be optimal, leading to a contradiction.

B.2.4 Proof of Theorem 2

Let $\bar{C}(\underline{\beta})$ denote the constraint set of the relaxed problem

$$\max_{x_L, x_\Delta \geq 0} \quad v(x_L, x_\Delta) := \frac{1}{1+l_1}(\underline{\beta} + x_\Delta)\frac{\bar{\beta}l_1 - x_L}{\bar{\beta}x_\Delta - \underline{\beta}x_L}(u_{H0} + \frac{\bar{u}_L x_L}{1+x_L} + \frac{(\bar{u}_H - \bar{u}_L)x_\Delta}{1+x_\Delta})$$
$$\text{s.t. } x_\Delta \geq x_L, \bar{\beta}x_L \geq \underline{\beta}x_\Delta, x_\Delta \geq l_1\underline{\beta} \text{ and } l_1\bar{\beta} \geq x_L \tag{13}$$

and $M(\underline{\beta})$ the set of optimal solutions to the problem. Define $\underline{\beta}_0 := \sup\{\underline{\beta} | M(\underline{\beta}) \subset \{\mathbf{x} | \frac{\bar{\beta}}{\underline{\beta}}x_L = x_\Delta\}\}$. We want to show that $M(\underline{\beta}) \subset \{\mathbf{x} | \frac{\bar{\beta}}{\underline{\beta}}x_L = x_\Delta\}$ for all $\underline{\beta} < \underline{\beta}_0$.

Suppose by contradiction that there exists $\underline{\beta} < \underline{\beta}_0$ such that a single-premium menu is optimal. Define $\underline{\beta}_1 := \sup\{\underline{\beta} < \underline{\beta}_0 | M(\underline{\beta}) \cap \{\mathbf{x} | x_L = x_\Delta\} \neq \emptyset\}$. By the maximum theorem, M is non-empty and upper-hemicontinuous. Thus, both a two-version menu and a single-premium menu will be optimal at $\underline{\beta}_1$. Let $\mathbf{x}^{2S}$ be the optimal two-version menu and $\mathbf{x}^1$ be the optimal single-premium menu at $\underline{\beta}_1$. By Lemma 3, $x_\Delta^{2S} > x_\Delta^1$.

Meanwhile, the definitions of $\underline{\beta}_1$ and $\underline{\beta}_0$ implies that only two-version menus can be optimal for $\underline{\beta} \in (\underline{\beta}_1, \underline{\beta}_0]$. Let $\mathbf{x}^{2S}(\underline{\beta})$ denote the optimal two-version menu there. Again, the maximum theorem implies that $x_\Delta^{2S}(\underline{\beta})$ is continuous in $\underline{\beta}$ and thus we can find a small enough $\delta > 0$ such that $x_\Delta^{2S}(\underline{\beta}_1 + \delta) > x_\Delta^1$.

Note that $\frac{\partial \ln v}{\partial \underline{\beta}}(\mathbf{x}^{2S}(\underline{\beta}_1 + \delta), \underline{\beta}) = \frac{1}{x_\Delta^{2S}(\underline{\beta}_1 + \delta) + \underline{\beta}} + \frac{1}{\bar{\beta}\frac{\bar{\beta}}{\underline{\beta}} - \underline{\beta}} < \frac{1}{x_\Delta^1 + \underline{\beta}} + \frac{1}{\bar{\beta} - \underline{\beta}} = \frac{\partial \ln v}{\partial \underline{\beta}}(\mathbf{x}^1, \underline{\beta})$ for any $\underline{\beta} \in (0, \bar{\beta})$. Hence, $\ln v(\mathbf{x}^{2S}(\underline{\beta}_1 + \delta), \underline{\beta}_1 + \delta) - \ln v(\mathbf{x}^{2S}(\underline{\beta}_1 + \delta), \underline{\beta}_1) < \ln v(\mathbf{x}^1, \underline{\beta}_1 + \delta) - \ln v(\mathbf{x}^1, \underline{\beta}_1)$. Rearrange the inequality and we have

$$\ln v(\mathbf{x}^{2S}(\underline{\beta}_1 + \delta), \underline{\beta}_1) - \ln v(\mathbf{x}^1, \underline{\beta}_1) > \ln v(\mathbf{x}^{2S}(\underline{\beta}_1 + \delta), \underline{\beta}_1 + \delta) - \ln v(\mathbf{x}^1, \underline{\beta}_1 + \delta) \geq 0.$$

The second inequality follows from the optimality of $\mathbf{x}^{2S}(\underline{\beta}_1 + \delta)$ at $\underline{\beta}_1 + \delta$. However, the constraint set expands as $\underline{\beta}$ decreases. So $x_\Delta^{2S}(\underline{\beta}_1 + \delta) \in \bar{C}(\underline{\beta}_1)$. The optimality of $\mathbf{x}^1$ then implies $\ln v(\mathbf{x}^{2S}(\underline{\beta}_1 + \delta), \underline{\beta}_1) - \ln v(\mathbf{x}^1, \underline{\beta}_1) \leq 0$, which leads to a contradiction.

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